1 **(a)** Define *moment of a force*.

The moment of a force *F* about an axis is the **product** of that force and

the **perpendicular** distance from the **line of action** of the force to the axis.

[1]

(b) A person supports a load of 20 N in his hand as shown in Fig 1.1. The system of the hand and load is represented by Fig 1.2. The rod represents the forearm and *T* represents the tension exerted in the biceps. The forearm weighs 65 N.

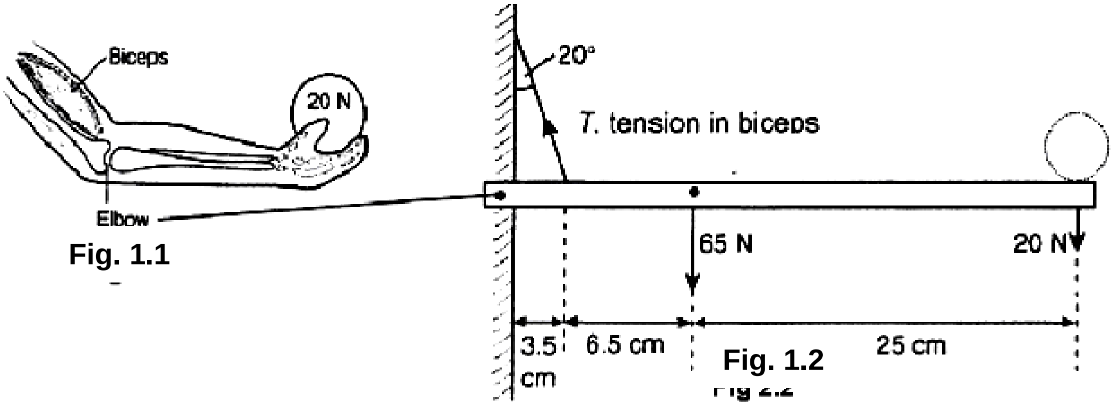


Fig. 1.1

Fig. 1.2

(i) Show that the tension *T* in the biceps is 410 N.

Taking moments about the elbow,

$T_y \times 3.5 = 65 \times 10 + 20 \times 35$

$T_y = 385.7$ N

$T = T_y / \cos 20° = 410$ N

[3]

(ii) Determine the magnitude and direction of the force acting at the elbow.

$T_x = 385.7 / \tan 20° = 140.4$ N

$\sum F_x = 0$

$T_x = R_x = 140.4$ N

$\sum F_y = 0$

$T_y = R_y + 65 + 20$

$R_y = 385.7 - 65 - 20 = 300.7$ N

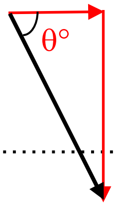
Resultant force acting at elbow (pivot)

$= \sqrt{R_x^2 + R_y^2} = 332$ N

State direction

$\tan \theta° = (300.7 / 140.4)$

$\theta° =$



force acting at the elbow = ... N

direction of the force: .... [4]

2 Fig. 2 shows a skateboarder of mass 55 kg about to descend a curved ramp in a skate